Electromyography as an Interaction Medium:

A Mini Literature Review

Draft

1. Introduction

Electromyography (EMG) measures the electrical activity generated by skeletal muscle contractions and has attracted sustained research interest as a non-invasive input and sensing modality for human-computer interaction (HCI). Historically rooted in prosthetics and clinical rehabilitation, EMG has progressively expanded its reach as wearable sensor hardware has become more affordable and machine learning has matured as a signal decoding tool. The eight works reviewed here collectively span the full width of this trajectory: from principled design frameworks and signal processing surveys, to real-time force-aware virtual reality interfaces, accessible facial and driving-oriented control systems, deep learning taxonomies, and critical design theory. Taken together, they illuminate both the meaningful technical progress the field has achieved and the considerable methodological, evaluative, and conceptual work that remains ahead.

2. Background and Scope

The physiological basis of EMG as a control signal is well established. When motor units fire, action potentials propagate along muscle fibres and summate into a surface-accessible bioelectric signal. Eddy, Scheme, and Bateman (2023) trace the historical arc from early one-muscle-one-function prosthesis control schemes through to pattern recognition-based myoelectric control, which uses machine learning to decode more nuanced, multi-channel EMG patterns. This history is directly relevant to HCI because, as the authors argue, the community largely inherited its methods from prosthetics without sufficiently interrogating whether those methods are appropriate for general-purpose interaction contexts. General-purpose HCI often requires discrete event detection, robustness to untrained users in everyday settings, and the ability to compete with more convenient alternatives such as touchscreens or voice interfaces; none of these demands were central concerns in the clinical prosthetics literature.

The comprehensive fatigue monitoring survey (Zia ur Rehman et al., 2023) extends the scope of EMG beyond command decoding to a sensing and physiological monitoring function, situating sEMG alongside mechanomyography (MMG), near-infrared spectroscopy (NIRS), and ultrasound as complementary windows into muscle state. This multi-modal perspective carries practical implications for HCI: muscle fatigue degrades EMG signal quality and classifier reliability over time, meaning any deployment of EMG-based input in sustained or physically demanding contexts must account for this degradation. The deep learning survey (Côté-Allard et al., 2020) and the feature optimization study (Khokhar et al., 2010) address complementary aspects of the signal processing pipeline; the former examines how neural

architectures can replace hand-engineered features, while the latter demonstrates how deliberate feature selection reduces computational overhead without sacrificing accuracy.

At the applied end of the spectrum, Zhang et al. (2022) demonstrate a real-time force-aware VR interface, while the facial EMG paper (Hamedi et al., 2014) and the driving feasibility study (Bi et al., 2019) each extend the application surface of EMG to non-hand-based and high-stakes interaction contexts respectively. The Research through Design (RtD) work on data-enabled digital musical instruments (DMIs) occupies a different register entirely, asking critical questions about what it means to encode biological phenomena as data and what kinds of interaction that encoding enables or forecloses.

3. Thematic Analysis

3.1 The Design Problem: From Prosthetics to General Purpose Use

A central argument in Eddy et al. (2023) is that EMG-based HCI research has been hampered by the uncritical adoption of methods designed for a fundamentally different use case. The authors characterise four recurring failure modes across approximately fifty sampled ACM papers: insufficient attention to interaction design (particularly the distinction between static contractions and dynamic gestures); sub-optimal model design borrowed from prosthetics without adaptation; over-reliance on offline evaluation metrics that correlate poorly with real-world usability; and poor reproducibility. Their proposed eight-step design framework, spanning input type selection, technology choice, control scheme, algorithm selection, recognition paradigm, parameter tuning, model training, and evaluation, offers a structured corrective to these patterns.

Zhang et al. (2022) can be read as a partial instantiation of the principled design process Eddy et al. advocate. The force-aware VR interface is built around a clearly articulated interaction need (natural, unobtrusive force input for immersive environments), makes deliberate technology choices (an off-the-shelf 8-channel sEMG wristband at 2000 Hz, a frequency-domain CNN, and Unity integration), and evaluates the system both objectively and through a psychophysical user study. Critically, the system demonstrates transfer to new users with under two minutes of calibration, addressing the generalisation challenge that Eddy et al. identify as one of the most pressing open problems in the field.

The facial EMG interface (Hamedi et al., 2014) and the driving feasibility study (Bi et al., 2019) push the scope of EMG-based HCI in directions that the prosthetics literature offers even less guidance on. Facial EMG repurposes muscles not typically implicated in motor control research as intentional input channels, supporting hands-free interaction for users with limited upper-limb mobility. The driving study relocates EMG from controlled gesture recognition tasks to a continuous, safety-critical, high-cognitive-load scenario, finding that participants could navigate simulated driving environments using EMG-decoded wrist and arm signals. Both works illustrate that the problem space for EMG in HCI is considerably wider than the prosthetics heritage might suggest, and that each new application context carries its own design requirements that cannot simply be read off from prior work.

3.2 Signal Processing and Modelling Approaches

A consistent thread across the corpus is the question of how best to represent and learn from raw EMG signals. The fatigue survey reviews a wide range of analysis techniques, including FFT, STFT, continuous wavelet transforms, entropy measures, PCA, ICA, and a variety of machine learning classifiers (CNNs, SVMs, LSTMs, KNN, LDA, and Random Forests), noting that sEMG remains susceptible to noise, cross-talk, and non-stationarity during dynamic contractions. These are not merely theoretical concerns; they translate directly into reduced classification reliability in any deployed system.

The feature optimization study (Khokhar et al., 2010) addresses this challenge at the feature engineering level, evaluating 42 candidate features drawn from the time domain, frequency domain, and time-frequency representations of forearm EMG. The study demonstrates that a principled feature selection process can significantly reduce computational cost while preserving, and in some cases improving, gesture recognition accuracy across four wrist movement classes. This finding is practically important for real-time wearable systems where processing overhead is a genuine constraint, and it provides a useful counterpoint to the assumption in the deep learning survey that more parameters are always preferable.

The deep learning survey (Côté-Allard et al., 2020) situates the feature engineering question within a broader architectural shift, arguing that CNNs, RNNs, LSTMs, autoencoders, and hybrid CNN-RNN models can learn richer signal representations than hand-crafted features allow. By learning directly from raw or minimally processed EMG data, these architectures reduce the domain knowledge required for feature design and improve robustness to inter-user and inter-session variability. The survey also notes, however, that deep models are data-hungry; the small datasets typical of EMG research constrain both training quality and generalisation to unseen users.

Zhang et al. (2022) offer the most methodologically distinctive approach to signal representation in this corpus. Rather than working in the time domain, they apply Short-Time Fourier Transform (STFT) to convert raw EMG into spectrograms, treating each electrode channel as an image dimension and exploiting a lightweight CNN encoder-decoder. Their joint classification-regression loss, which determines whether force is being applied before estimating its magnitude, addresses the sparsity and high variance of real-world interaction forces. The result is a 3.3% mean force estimation error with 624 ms of input history, at a computational cost low enough for real-time inference. The frequency-domain approach also offers practical advantages in noise isolation relative to time-domain feature methods, making it a promising candidate for adoption in other EMG decoding tasks.

3.3 Evaluation, Generalisability, and Reproducibility

Evaluation rigour is a persistent concern across the reviewed works. Eddy et al. (2023) note that roughly half of the HCI EMG papers they sampled relied exclusively on offline classification accuracy, a metric with well-documented poor correlation to online usability. The fatigue survey raises analogous concerns: the majority of reviewed studies were short-term and lab-based, with typical participant samples of five to twenty individuals, severely limiting claims about real-world applicability. The feature optimization study

and the facial EMG paper share similar constraints, both working with small participant counts and limited gesture vocabularies in tightly controlled laboratory conditions.

Zhang et al. (2022) are more rigorous on evaluation than much of the reviewed literature, conducting both objective model assessment and a psychophysical user study that measures perceptual benefit in VR. Their tapping detection experiment is a notable exception, however, having been conducted on a single subject; the authors acknowledge this limitation directly. The driving feasibility study (Bi et al., 2019) is notable for moving into a higher-stakes, higher-complexity evaluation context than most EMG-HCI work attempts, though it too remains simulation-based and involves a small number of participants.

Reproducibility receives particular attention from Eddy et al. (2023), who document inconsistent terminology, undefined parameters, custom hardware, and absent datasets or source code throughout the sampled literature. This concern is directly answered by Zhang et al., who release both their EMG-Force dataset and model code publicly. The fatigue survey similarly identifies the need for larger, open-source, multi-site datasets as a priority for the field. Across the corpus as a whole, infrastructure, not just algorithms, emerges as a key bottleneck for progress in EMG-based HCI research.

3.4 Expanding the Application Surface

Several papers in this corpus extend EMG-based interaction into domains that the mainstream prosthetics and gesture recognition literature has not prioritised. The facial EMG interface (Hamedi et al., 2014) demonstrates that facial muscle activity, detected via surface electrodes and classified in real time, can support computer control for users with limited hand or arm mobility. The system translates eyebrow movements and jaw activations into interface commands, with real-time audiovisual feedback supporting user learning and reducing the burden of interpreting system responses. While the current gesture vocabulary is limited and electrode placement sensitivity remains a practical challenge, the work establishes facial EMG as a viable modality for accessible, hands-free interaction.

The driving feasibility study (Bi et al., 2019) relocates EMG to a continuous, safety-critical context where the interaction demands are qualitatively different from the discrete gesture recognition tasks most systems are designed for. Participants successfully navigated simulated driving scenarios using EMG signals decoded from wrist and arm muscles, and the study measured cognitive workload and usability ratings alongside control accuracy. The finding that EMG interfaces can support complex, time-pressured tasks expands the plausible application space considerably, particularly for users with upper-limb impairments for whom conventional vehicle controls may be inaccessible. Both the facial and driving applications reinforce Eddy et al.'s broader point that generalising beyond prosthetics requires thinking carefully about the specific demands of each target context, rather than transplanting methods wholesale from one setting to another.

3.5 The Philosophical Dimension: Data, Ambiguity, and Apparatus

The RtD paper on DMIs introduces a dimension absent from the other seven works: a critical interrogation of what it means, at a foundational level, to design with biological data. Drawing on Karen Barad's concept of the apparatus, in which measurement does not passively record a pre-existing phenomenon but actively co-constitutes it, the authors propose the notion of the "ambiguity shift" to describe how data-enabled artefacts reconfigure, rather than eliminate, ambiguity. This framing has direct relevance to EMG-based interfaces: when a system decodes muscle activity as a command, it is not transparently reading a pre-formed user intention but establishing, through its model architecture and learned mappings, what counts as a valid input and what gets discarded.

The VoxEMG case study, which uses sEMG to sonify vocal muscle tension as an expressive instrument rather than a command channel, exemplifies a design stance that productively resists the reduction of biological signal to discrete control output. This contrasts with the optimisation-centred framing dominant in the rest of the corpus, where ambiguity is treated primarily as noise to be minimised through better features, more robust classifiers, or open-set recognition schemes. The RtD perspective does not invalidate the engineering agenda; robustness and low latency remain genuine requirements for deployed systems. It does, however, invite the community to reflect on what values are encoded in the way we model human physiology as data, and what kinds of interaction those modelling choices make possible or foreclose. The question of whether AI and ML latent spaces can support productive indeterminacy without reintroducing explicit concepts through loss function design is one that bridges both the philosophical and technical literatures.

4. Cross-Cutting Themes and Gaps

Several open problems recur across multiple works in this corpus. User and context independence, understood as the ability to deploy EMG systems without per-user retraining or environmental recalibration, is identified by Eddy et al. (2023) as critical for consumer adoption, partially addressed by Zhang et al.'s calibration protocol, and implicitly at stake in the fatigue survey's concern with generalising across contraction types and populations. The deep learning survey identifies the same problem from an architectural angle, noting that cross-user and cross-session generalisation remains one of the central unsolved challenges for neural EMG models. No single work in this corpus fully resolves the tension between model personalisation, which improves performance, and model generality, which improves usability.

Activity detection in discrete control scenarios, specifically the reliable segmentation of gesture onset and offset from a continuous EMG stream, is highlighted by Eddy et al. as under-researched and is not substantively addressed by any other work in this corpus. This is a practical gap with real consequences: most deployed systems would encounter continuous, unconstrained EMG rather than the pre-segmented trials typical of laboratory evaluations. The feature optimization study, the deep learning survey, and the driving paper all work primarily with pre-segmented or simplified input assumptions, and the general problem of activity detection in noisy, real-world streams remains open.

Multi-modal sensing is another recurrent recommendation across the corpus. The fatigue survey argues that combining sEMG with NIRS and MMG improves fatigue classification accuracy and robustness.

Eddy et al. advocate for EMG-plus-IMU integration to improve context independence. Zhang et al. note that combining EMG with hand tracking data could help differentiate active and resisting forces. These recommendations converge on a shared intuition: no single sensing modality captures the full picture of human physical intent, and meaningful progress may require treating sensor fusion as a first-class design concern rather than an optional enhancement.

Accessibility is a notable gap in the corpus considered as a whole. The facial EMG and driving papers are most directly concerned with users who cannot rely on conventional hand-based interfaces, and Zhang et al. mention accessibility as a future direction. Yet the bulk of the reviewed work focuses on able-bodied participants performing predefined gesture vocabularies in controlled settings. Given that EMG-based control originated in prosthetics precisely because it provides a natural interface for people with absent or impaired motor pathways, the relatively thin coverage of disability and inclusion in general-purpose EMG-HCI research represents a significant missed opportunity.

Finally, the DMI paper raises a question that most works in this corpus leave implicit: for whom are these systems being designed, and to what ends? Engineering robustness and reducing calibration burden are genuinely important goals, but they are not neutral. The choice to optimise for speed and accuracy, for gesture vocabulary size, or for plug-and-play deployment reflects particular assumptions about who the user is and what they want to do. Broadening the participant populations studied, the interaction contexts targeted, and the design values prioritised may be as important to the field's long-term progress as any advance in signal processing or network architecture.

5. Conclusion

The eight works reviewed here collectively map the current terrain of EMG-based interaction research across several registers: principled framework-building, multi-modal fatigue sensing, feature engineering and deep learning, real-time force-aware VR, accessible and high-stakes application contexts, and critical design theory. The picture that emerges is of a field that has made meaningful technical progress, particularly in machine learning-based decoding, wearable hardware accessibility, and the expansion of interaction scenarios, but that continues to grapple with persistent challenges around evaluation rigour, reproducibility, dataset scale, and generalisation. The critical design perspective introduced by the DMI paper serves as a productive reminder that technical robustness and conceptual richness are not competing values, and that how we model human physiology as data carries consequences that extend well beyond classification accuracy metrics.

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Paper Individual Findings Data

Non-invasive Techniques for Muscle Fatigue Monitoring: A Comprehensive Survey:

- **Purpose:** To survey non-invasive techniques for muscle fatigue monitoring and detection, covering the principles, mechanisms, parameters, applications, and tradeoffs of EMG, MMG, NIRS, and ultrasound.
- **Tech Used** Surface EMG (sEMG) and intramuscular EMG (iEMG), mechanomyography (MMG), near-infrared spectroscopy (NIRS, including CW-NIRS, TD-NIRS, FD-NIRS), and ultrasound (A-mode, B-mode, E-mode/elastography). Signal analysis methods include FFT,

STFT, CWT, wavelet packet analysis, entropy measures, PCA, ICA, and ML classifiers (CNN, SVM, LSTM, KNN, LDA, RF).

- **Methodologies:** PRISMA-based systematic review of IEEE Xplore, PubMed, and Scopus, covering all years up to 2023. Initial pool of 3,502 papers reduced to 122 after deduplication, title/abstract screening, and full-text review.
- **Contributions:** Comprehensive comparison of four non-invasive modalities across isometric, dynamic, and electrically-evoked contraction types. Includes a summary table of parameters, strengths, and limitations per modality. Also covers multi-modal fusion approaches (e.g., sEMG+NIRS+MMG) and their accuracy improvements.
- **Limitations:** Most reviewed studies were short-term and lab-based, limiting real-world applicability. Small sample sizes (typically 5–20 participants). Single-modality studies dominate despite known reliability issues. sEMG is susceptible to noise, cross-talk, and non-stationarity during dynamic contractions. NIRS has a temporal lag relative to EMG/MMG. The US cannot capture the full muscle in its field of view.
- **Open Problems Identified:** Need for larger, open-source, high-quality multi-site datasets. Real-time HD-sEMG decomposition remains immature. Multi-sensor miniaturization without data quality loss. More robust algorithms for dynamic, non-stationary signals. Better fatigue compensation methods for myoelectric control systems. Expanded study of multi-modal fusion for improved classification accuracy.

Shifting Ambiguity, Collapsing Indeterminacy: Designing with Data as Baradian Apparatus:

- **Purpose: To examine** how designing data-enabled artefacts inevitably reconfigures patterns of ambiguity, using Karen Barad's agential realism as a theoretical lens. The paper argues that data-enabled artefacts act as Baradian apparatuses that co-produce rather than neutrally measure phenomena, and proposes the concept of the "ambiguity shift" to describe how this reconfiguration operates in practice.
- **Tech Used:** Three DMI case studies: the VoxEMG (sEMG-based vocal muscle sensing platform using differential amplification, Bela embedded computing, and white-noise sonification), FM Tone Transfer (DDSP-based neural network audio plugin using pitch/loudness extraction and FM synthesis), and the (Un)ambiguous Guitar (augmented electric guitar with TMR magnetic bend sensors, hexaphonic pickup, pitch detection, transient modelling, and three Bela processing units).
- **Methodologies:** Research through Design (RtD), autoethnography, qualitative user studies with expert performers, and post-hoc diffractive analysis of three existing design artefacts. Theoretical framing draws on Barad's agential realism, Gaver et al.'s ambiguity taxonomy, and Giaccardi's three shifts in data-enabled RtD.
- **Contributions:** Introduces the "ambiguity shift" as a fourth type of shift in data-enabled RtD, alongside Giaccardi's agential, temporal, and infrastructural shifts. Proposes two mechanisms by which this shift operates: explicit/implicit concept encoding, and the Baradian argument that measurement co-determines phenomenon. Applies these frameworks concretely to three DMI

case studies with a unified reflective question set. Offers actionable design implications, including leaving things deliberately indeterminate and resisting over-conceptualization.

- **Limitations:** The three case studies arise from different musical backgrounds and research focuses, producing natural disparity in narration and reflection. The work is reflective rather than methodological and does not claim to generalize beyond the particulars examined. The authors acknowledge their own entanglement as participant-observers, meaning no neutral vantage point exists for the analysis.
- **Open Problems Identified:** How to design for productive indeterminacy without resulting in uninteresting or confusing interactions. How to evaluate or communicate the value of artefacts that deliberately avoid encoding well-defined conceptual knowledge. How to prevent explicit concepts from dominating design attention in ways that foreclose richer possibilities. Whether and how AI/ML latent spaces can support indeterminacy without reintroducing explicit concepts through loss functions or training objectives.

A Framework and Call to Action for the Future Development of EMG-Based Input in HCI:

- **Purpose:** To characterize the limitations of existing EMG-based input research in HCI, propose a structured design framework for building EMG-based control systems for general-purpose applications, and provide a research agenda (call to action) for the HCI community to unlock the broader potential of EMG beyond prosthesis control.
- **Tech Used:** Surface EMG (sEMG) sensors, including the Myo Armband (8-channel, 200 Hz, BLE) as a representative commercial wearable. References to IMUs (inertial measurement units) as complementary sensors. No new hardware was introduced; the paper is a literature synthesis and framework proposal.
- **Methodologies:** Non-systematic literature sampling of approximately 50 EMG-based HCI papers from the ACM Digital Library. Categorical data extraction from each paper covering: application/context, inputs, algorithms, features, windowing technique, training methods, user/context dependence, evaluation techniques and metrics, reported accuracy, participant count, and technological setup. Findings were synthesized through the interdisciplinary experience of a combined HCI and biomedical engineering/prosthetics research team.
- **Contributions**
    1. Identification and characterization of four recurring limiting themes in HCI EMG research: interaction design, model design, system evaluation, and reproducibility.
    2. An 8-step design framework organized into three categories (Interaction Design → Model Design → System Evaluation), covering: input type selection, technology selection, control scheme, algorithm selection, recognition type, parameter selection, model training, and evaluation.
    3. A call to action and research agenda identifying key open directions for EMG-based HCI, including discrete vs. continuous input distinction, wearable form factors, plug-and-play toolkits, and user/context-independent models.

4. Two worked framework examples (RC car controller, music control during exercise) demonstrate how the framework guides design decisions.

- **Limitations:** The paper's own literature sampling was non-systematic and focused exclusively on the ACM Digital Library, excluding IEEE and clinical venues. Paper selection was heavily informed by the authors' own research group experience, introducing potential selection bias. The framework does not propose new technical steps but rather organizes and translates existing best practices from prosthetics into an HCI-oriented format. The framework does not comprehensively address all intricacies of EMG-based control (e.g., activity detection, adaptation, and signal quality are treated as supplementary considerations rather than framework steps).
- **Open Problems Identified**
    - Development of robust user-independent and context-independent EMG models for general-purpose use.
    - Optimal strategies for activity detection (segmenting dynamic gestures from continuous EMG streams) in online discrete control scenarios.
    - Identification of standardized, intuitive gesture sets for general-purpose HCI that minimize false activations during everyday activity.
    - Exploration of open-set recognition for EMG to improve robustness beyond closed-set assumptions.
    - Better understanding of discrete control parameter selection (window length, increment, feature sets) outside the prosthesis-control paradigm.
    - Development of validated, open-source EMG toolkits supporting both continuous and discrete control (extending beyond tools like EMBody).
    - Investigating combined EMG + IMU sensing for improved dynamic gesture recognition and context independence.
    - Gamified and low-burden user training and model adaptation strategies for general-purpose users.
    - Exploration of novel EMG form factors (watches, bracelets) and alternative sensing modalities (EMS, implanted electrodes) for HCI contexts.
    - Understanding the potential divergence of EMG-based HCI research from prosthetics-driven paradigms as a distinct research subfield.

Force-Aware Interface via Electromyography for Natural VR/AR Interaction:

- **Purpose:** To design and evaluate a real-time, learning-based neural interface that decodes finger-wise interaction forces from forearm surface EMG signals and applies them to virtual objects in VR/AR environments, closing the physicality gap between real-world manual dexterity and immersive virtual interaction without requiring obtrusive on-hand sensors.
- **Tech Used** Off-the-shelf wireless surface EMG sensors (8-channel forearm wristband, 2000 Hz sampling rate), MorphSense force sensing trackpad (~125 Hz) for ground truth force data collection, short-time Fourier transform (STFT) for frequency-domain signal transformation, a custom lightweight CNN with encoder-decoder architecture and bottleneck design, Unity-based

VR environment for interaction demonstrations, and standard VR/AR display hardware. Dataset and source code released publicly at https://github.com/NYU-ICL/xr-emg-force-interface.

- **Methodologies:** Custom EMG-Force dataset collection from 9 participants (ages 21–30, 5 female, 4 male) across 11 hand-object interaction types (pressing and pinching variants), yielding 990 seconds of time-synchronized EMG and force data per participant. Frequency-domain CNN trained on STFT spectrograms of raw EMG using a joint classification-regression loss (cross-entropy for force/no-force detection, L2 for force magnitude regression). Generalization to new users was evaluated via a short (~1-minute) calibration protocol. Robustness tested across sensor placement shifts, task context changes, and inter-subject variation. User study (psychophysical experiment) assessing whether force-aware feedback enhanced perception of virtual object material properties (stiffness discrimination). Separate finger tapping detection experiment using a single subject with 1200 seconds of data, evaluated using precision and recall with temporal IoU thresholding.
- **Contributions**
    1. First end-to-end, real-time EMG-based neural interface for decoding continuous, simultaneous, finger-wise interaction forces for VR/AR, achieving 3.3% mean error.
    2. A novel joint classification-regression loss function that handles the sparse and high-variance nature of human-applied forces, reducing false positives and false negatives relative to naive regression baselines.
    3. A custom EMG-Force dataset covering 11 common multi-finger pressing and pinching interactions with time-synchronized EMG and force ground truth, released publicly.
    4. A frequency-domain (STFT spectrogram) input representation for EMG that improves noise isolation and leverages CNN architectures more efficiently than time-domain approaches.
    5. Demonstration of practical generalizability to new users with under 2 minutes of calibration, validated across sensor placement and user variation.
    6. A psychophysical user study confirming that force-aware VR interaction significantly enhances perception of virtual object physical properties (stiffness).
    7. Proof-of-concept for ubiquitous finger-tap detection (92.0% precision, 92.7% recall across all 10 fingers) as an eyes-free interaction modality.
- **Limitations:** The dataset was collected using a flat pad-like force sensor, meaning the model does not robustly generalize to complex 3D object geometries or higher-DOF hand configurations beyond pressing and pinching on flat surfaces. A short calibration step (~1 minute) is still required for new users, limiting true plug-and-play deployment. The model may conflate active (volitional, unconstrained) forces with resisting forces (opposing a physical object), as these can produce different EMG patterns. The tapping detection experiment was conducted on only one subject, limiting the generalizability of those results. The system has not been evaluated on populations with limb impairments or atypical musculature.
- **Open Problems Identified**
    - Extension to free-form 3D object interaction beyond flat-surface pressing and pinching, potentially requiring more advanced wearable sensing devices.
    - Development of unsupervised or self-supervised calibration to eliminate the need for any per-user setup, enabling fully adaptive zero-shot generalization.

- Differentiating active volitional forces from resisting/reactive forces, potentially through integration of hand tracking and arm pose estimation.
- Broader accessibility evaluation, including users with limb impairments or motor differences, where force-aware VR/AR could be especially impactful.
- Scaling the approach to whole-body or higher-DOF interactions that go beyond finger-level force decoding.
- Combining EMG-based force decoding with haptic output (EMS/haptic rendering) to fully close the bidirectional interaction loop in VR/AR.

Deep Learning for EMG-based Human-Machine Interaction: A Review:

- **Purpose:** This paper surveys how deep learning techniques are being applied to electromyography (EMG) signals for human–machine interaction systems, with a focus on replacing traditional hand-engineered features with neural network models.
- **Tech used:** The review discusses several deep learning architectures used for EMG decoding:
    - Convolutional Neural Networks (CNNs)
    - Recurrent Neural Networks (RNNs)
    - Long Short-Term Memory networks (LSTMs)
    - Autoencoders
    - Hybrid CNN-RNN architectures

    These models are applied to gesture recognition, prosthetic control, rehabilitation systems, and wearable interfaces.

- **Methodologies:** The authors perform a literature survey of deep learning EMG systems, analyzing how neural networks are used for signal feature extraction, temporal modelling of muscle activation and cross-user generalization. They compare deep learning models with classical approaches like SVM, LDA, and KNN classifiers.
- **Contributions:**
    1. A taxonomy of deep learning approaches for EMG-based interaction
    2. Identification of how neural networks reduce reliance on handcrafted features
    3. Comparison of different network architectures and their performance
    4. Discussion of real-time wearable HCI applications

    The review argues that deep learning allows systems to **learn richer representations of EMG signals**, improving gesture classification and robustness to noise.

- **Limitations:** The paper highlights several limitations in current EMG-DL research:
    - Small datasets limit model generalization
    - Many studies remain lab-based rather than real-world deployments
    - Training deep networks requires large annotated EMG datasets

A Novel Feature Optimization for Wearable Human-Computer Interfaces Using Surface Electromyography Sensors:

- **Purpose:** The study investigates feature optimization methods for EMG-based wearable human-computer interfaces, aiming to improve gesture recognition accuracy while reducing computational complexity.
- **Tech Used:** The system includes:
    - surface EMG sensors placed on the forearm
    - signal processing pipelines extracting 42 different features
    - machine learning classifiers for gesture recognition
  - Features are drawn from:
    - time domain signals
    - frequency domain signals
    - time-frequency representations
- **Methodologies:** Researchers recorded EMG signals from four wrist movement classes using forearm electrodes. The workflow included EMG signal acquisition, feature extraction, feature selection optimization, and classifier training and testing.

  The authors propose an optimized feature selection method that removes redundant features while preserving classification accuracy.

- **Contributions:** The study demonstrates that:
  1. Feature selection can significantly reduce computational cost
  2. Optimized feature sets improve gesture recognition accuracy
  3. Efficient feature pipelines are critical for real-time wearable EMG interfaces

  The work also highlights how feature engineering plays a major role in EMG-based interaction systems.

- **Limitations:**
  - The experiment involved a limited number of gesture classes
  - Small participant sample sizes
  - Focused only on wrist movements, limiting generalization to more complex gestures

Face-Computer Interface: Intent Recognition Based on Facial Electromyography:

- **Purpose:** This paper introduces a facial electromyography (fEMG)-based human-computer interface designed to recognize user intent from facial muscle activity and translate it into computer commands.
- **Tech used:** The system combines:
  - Facial EMG sensors
  - Machine learning classifiers
  - Real-time audiovisual feedback interfaces

The system detects facial muscle activation patterns corresponding to user commands.

- **Methodologies:** Participants performed specific facial gestures, such as eyebrow movements and jaw activity. The system records facial EMG signals, extracts signal features, classifies the gestures using machine learning models, and translates recognized gestures into interface commands. The interface also provides real-time visual and audio feedback to improve user learning and interaction.
- **Contributions:**
    1. Demonstration of facial EMG as a non-hand-based input modality
    2. Real-time intent recognition for computer interaction
    3. Exploration of accessible interfaces for users with limited hand mobility

  The authors show that facial EMG can support hands-free interaction scenarios.

- **Limitations:**
    - Facial EMG signals are sensitive to noise and electrode placement
    - Limited gesture vocabulary
    - Experiments were conducted in controlled laboratory conditions

Evaluating the Feasibility of EMG-Based Human–Machine Interfaces for Driving:

- **Purpose:** This research evaluates whether EMG-based human–machine interfaces can support complex real-world tasks such as driving, particularly for individuals with upper-limb impairments.
- **Tech used:**
    - Surface EMG sensors detecting arm and wrist muscle activity
    - EMG decoding algorithms translate signals into vehicle control commands
    - Usability evaluation frameworks measuring cognitive workload and safety
- **Methodologies:** Participants performed simulated driving tasks using an EMG-based control interface. Researchers measured task performance, control accuracy, cognitive workload, and usability ratings. The system's performance was compared to traditional control approaches.
- **Contributions:**
    1. EMG interfaces can enable alternative control pathways for complex tasks
    2. Users were able to successfully navigate simulated driving scenarios
    3. EMG control could improve accessibility for individuals with limb differences
    4. The study expands EMG-based HCI beyond simple gestures to high-stakes interaction environments.
- **Limitations:**
    - Controlled simulation environment rather than real-world driving
    - Small participant sample
    - EMG interfaces still require calibration and training